

Sion King

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EDUCATION

University of Central Florida

Bachelor of Science in Statistics - 3.4 GPA

Orlando, FL

Expected December 2027

EXPERIENCE

College of Sciences Student Ambassador

University of Central Florida

August 2026 – Present

Orlando, FL

- **Facilitated 8 interactive meetings** and social events per semester, engaging an average of **30 attendees** per event to **strengthen peer community** and campus belonging
- Served as a liaison between **16,000+ College of Sciences students** and university leadership, surfacing student feedback that **contributed to** revisions to **student programming** and **engagement initiatives**
- Cultivated relationships with **20+ faculty members** and **30+ alumni**, increasing participation in networking events and **expanding mentorship opportunities** for College of Sciences students

Finance Team @ KnightHacks UCF

University of Central Florida

July 2026 – Present

Orlando, FL

- Managed **budgeting** and **financial planning** for Knight Hacks, helping manage a **\$100K+ annual budget** across sponsorship revenue, event programming, and operating expenses
- Refined the **procurement pipeline** by standardizing purchase requests, approval tracking, and reimbursement documentation, **reducing** average processing time by **34 days**
- Tracked departmental spending across **6 teams** and provided recurring budget updates, helping leadership identify variances and **prevent cost overruns**

PROJECTS

Halo

- Won **1st of 77** projects for **Waymo** and **1st of 34** for **State Farm** at **ShellHacks 2026** with a blind-spot detection bike helmet, leading its computer vision, sensor fusion, and real-time visualization
- Processed **real-time ultrasonic sensor data** using **Python** to estimate vehicle distance, trajectory, passing clearance, and time-to-collision
- Applied **median filtering** and **temporal thresholds** to reduce measurement noise, **decreasing false hazard alerts by 60%** and improving system reliability in real-time detection scenarios

Happiness and Social Media Multiple Linear Regression Model

- Analyzed **500 observations** across **13 predictors** using **R** to model the relationship between social media use, behavioral factors, and self-reported happiness scores
- Validated a multiple linear regression model using **10-fold cross-validation** to assess predictive performance, achieving an $R^2 = 0.639$, $RMSE = 0.920$, and $MAE = 0.750$
- Applied **response transformation** and **model diagnostics** to improve R^2 from 0.639 to 0.665, identifying sleep quality, stress level, and age as the 3 statistically significant predictors

Graduate School Multiple Linear Regression Model

- Analyzed **400 graduate school admission records** across **6 predictors** using **R** to identify academic factors associated with admission probability
- Developed a multiple linear regression model achieving an adjusted R^2 of 0.796, explaining **79.8% of variation** in graduate school admission outcomes
- Constructed **95% confidence and prediction intervals** and compared full versus reduced models, identifying a **4-predictor model** that retained **99.9% of explanatory power** and highlighted GRE, TOEFL, LOR, and CGPA as key admission drivers

TECHNICAL SKILLS

Relevant Coursework: Stat I-III, Big Data Analytics, Foundations of Data Science and AI, Data Science I

Statistical Methods: Multiple Linear Regression, Statistical Inference, Hypothesis Testing, Confidence & Prediction Intervals, Correlation Analysis, Regression Diagnostics, Model Selection

Data Analysis: Data Cleaning, Data Visualization, Predictive Modeling, Model Evaluation

Languages: R, Python, SQL, C

Tools & Libraries: Excel, Pandas, NumPy, Matplotlib, RStudio, Jupyter Notebook, Git/GitHub